

Lec5

CSS Units

CSS Units are used to define the size of elements, text, margins, padding, widths, heights, and other properties in CSS.

There are two main types of CSS units:

1. Absolute Units
2. Relative Units

Absolute Units

These units have a fixed size and do not change based on screen size.

Unit	Description
px	Pixels
cm	Centimeters
mm	Millimeters
in	Inches
pt	Points (1pt = 1/72 inch)
pc	Picas (1pc = 12pt)

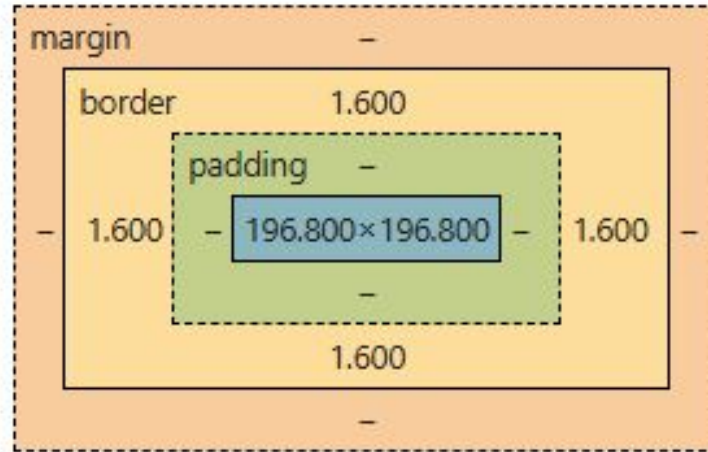
Relative Units

These units are relative to another value, making them useful for responsive design.

Unit	Description
%	Relative to parent element
em	Relative to parent element's font size
rem	Relative to root (<code>html</code>) font size
vw	1% of viewport width
vh	1% of viewport height
vmin	1% of smaller viewport dimension
vmax	1% of larger viewport dimension
ch	Width of the "0" character

Box Model

The **CSS Box Model** is a concept that describes how every HTML element is represented as a rectangular box. It determines the amount of space an element occupies on a web page.



1. Content

The actual content of the element, such as text, images, or videos.

```
div {  
  width: 200px;  
  height: 100px;  
}
```

Here, the content area is $200\text{px} \times 100\text{px}$.

2. Padding

The space between the content and the border.

```
div {  
  padding: 20px;  
}
```

Padding adds space inside the element around the content

3. Border

The line surrounding the padding and content.

```
div {  
  border: 2px solid black;  
}
```

4. Margin

The space outside the border that separates the element from other elements.

```
div {  
  margin: 30px;  
}
```

Complete Example

HTML

```
<div class="box">  
  Hello World  
</div>
```

CSS

```
.box {  
  width: 200px;  
  height: 100px;  
  padding: 20px;  
  border: 5px solid black;  
  margin: 30px;  
}
```

Pseudo Classes

A **Pseudo-Class** is used to style an element based on its current state or position.

Syntax

```
selector:pseudo-class {  
  property: value;  
}
```


Example

:hover

Applies styles when the mouse pointer is over an element.

```
button:hover {  
  background-color: blue;  
  color: white;  
}
```

:focus

Applies styles when an input field is selected.

```
input:focus {  
  border: 2px solid blue;  
}
```

:first-child

Selects the first child element.

```
li:first-child {  
  color: red;  
}
```

:last-child

```
li:last-child {  
  color: green;  
}
```

:nth-child()

```
li:nth-child(2) {  
  color: blue;  
}
```

:visited

```
a:visited {  
  color: purple;  
}
```

Pseudo Elements

A **Pseudo-Element** is used to style a specific part of an element.

Syntax

```
selector::pseudo-element {  
  property: value;  
}
```

Example

::first-letter

Styles the first letter of text.

```
p::first-letter {  
  font-size: 30px;  
  color: red;  
}
```

::first-line

Styles the first line of text.

```
p::first-line {  
  font-weight: bold;  
}
```

::before

Inserts content before an element.

```
p::before {  
  content: "👉";  
}
```

::after

Inserts content after an element.

```
p::after {  
  content: "✔";  
}
```

::selection

Styles the text selected by the user.

```
::selection {  
  background-color: yellow;  
  color: black;  
}
```

Difference Between Pseudo-Class and Pseudo-Element

Pseudo-Class	Pseudo-Element
Represents a special state of an element	Represents a specific part of an element
Uses <code>:</code>	Uses <code>::</code>
Example: <code>:hover</code> , <code>:focus</code> , <code>:first-child</code>	Example: <code>::before</code> , <code>::after</code> , <code>::first-letter</code>
Styles existing elements based on conditions	Styles or inserts content into parts of elements

Task

Create cards with hover effects and proper spacing

